Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Use figure 0.1 to complete the following.

a. Create a frequency distribution table from figure 0.1 (approximate).

hours	freq.

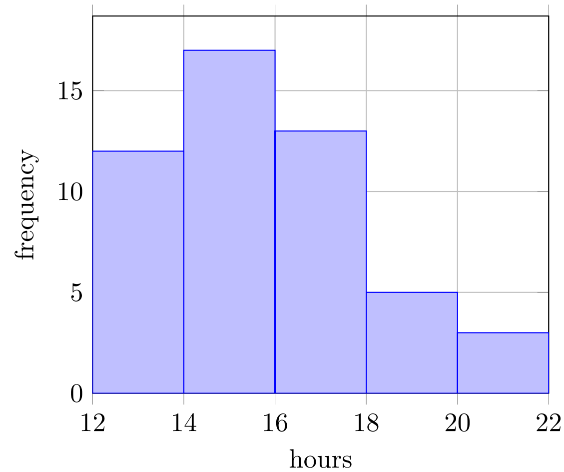

Figure 0.1: Credit hours of 50 students.

b. Using your frequency table, find the mean.

c. What percentage of students are taking less than 14 hours?

2. 10 randomly selected MHU students were surveyed to determine the number of parking tickets they received this semester. The following results were found:

0 7 0 5 1 4 1 2 1 2

a. Find the mean.

c. Find the mode.

b. Find the median.

d. Find the standard deviation..

3. Two math 107 sections take an exam, and the following is found. The 1^{st} section has a mean score of 75 with standard deviation 15, and the 2^{nd} section has a mean score of 80 with a standard deviation of 10.
Are the scores more consistent in the 1^{st} or 2^{nd} section? Why?

4. A population is normally distributed with mean 50 and standard deviation 15. Draw a sketch of the bell curve labeling the mean and the intervals representing 1, 2, and 3 standard deviations of the mean.

5. Find the following probabilities,

a. $p(z < .67)$

c. $p(.09 < z < .67)$

b. $p(z > .67)$

6. A population is normally distributed with a mean 100 and standard deviation 15. Find the following probabilities.

a. $p(x < 75)$

b. $p(75 < x < 115)$

7. Find c such that each of the following is true.

a. $p(z < c) = 9.18\%$

b. $p(z > c) = 40.95\%$

8. [7 points each] The average score on the ACT is 18 with a standard deviation of 6.

a. Find the probability that a random test score is below 12.

b. What score separates the **bottom** 48.01% from the rest of the population.

9. [BONUS] The *Central Limit Theorem* states that for large enough n, then the means of all samples of size n will be normally distributed. With this in mind, suppose we take a large, randomly selected sample of men and find that $\bar{x} = 70.5$ inches. If it is known that $\sigma = 1.5$ inches, how can we say that this value is within 3 inches of the actual mean height, μ, with 95.4% certainty?

Honor Statement

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University.

Signature:________________________________